

Table 2. Qwen-assisted BRCA marker programmes. Complete programmes are supplied in Supplementary Table S4.

Subtype	Positive markers	Negative markers
LumA	<i>ESR1, GATA3, MAPT, DACH1</i>	<i>TUBA1C, UBE2C</i>
Her2	<i>AKT1, MYCN, EPCAM, MMP1</i>	<i>ESR1, IGF1R, KRT14, RARG</i>
LumB	<i>ESR1, GATA3, PCLAF, BUB3</i>	<i>EGFR, KRT14, DEFB1</i>
Normal	<i>EGFR, TXNIP, KIT, JAG1</i>	<i>MCM2, CENPF, FOXM1</i>
Basal	<i>CDC20, AURKB, FOXM1, TOP2A</i>	<i>GATA3, NAT1, ELOVL5</i>